Minecraft Server Setup

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Information

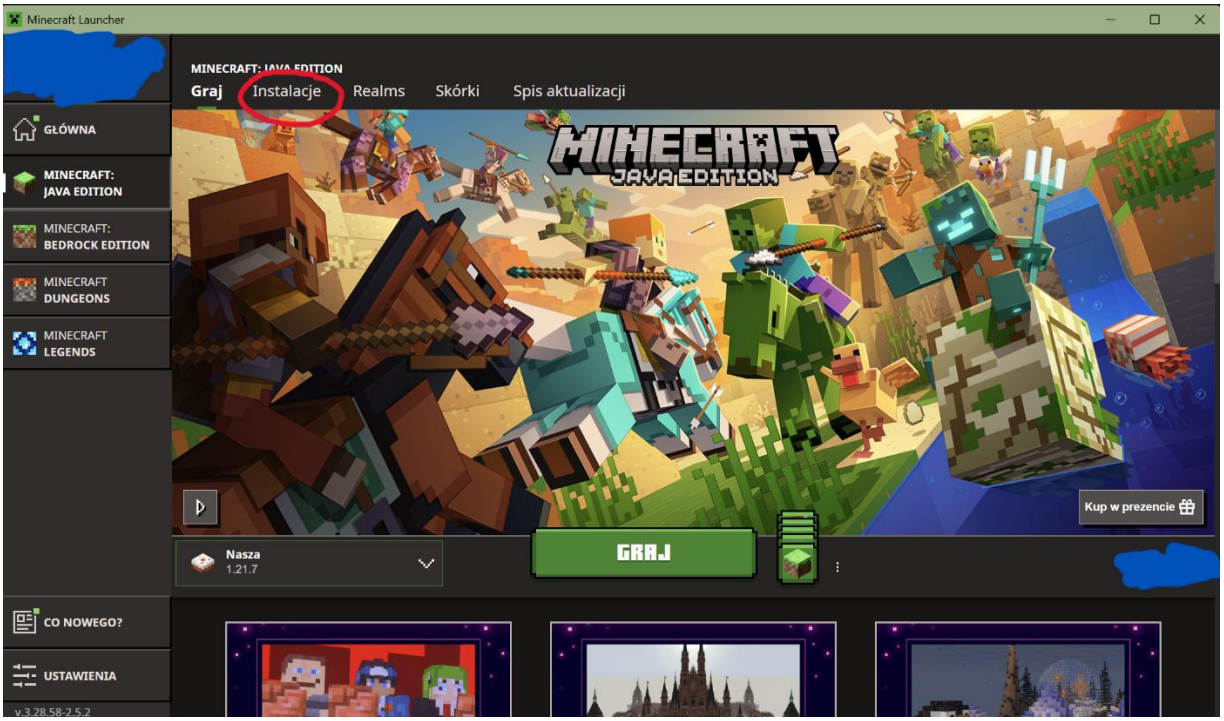
This documentation was prepared about a year after the server was created and on a different computer, therefore the screenshots included in it are intended solely to demonstrate familiarity with the performed steps.

Procedure

1. Downloading the package

To set up the server, the server.jar file is required. It can be downloaded as follows:

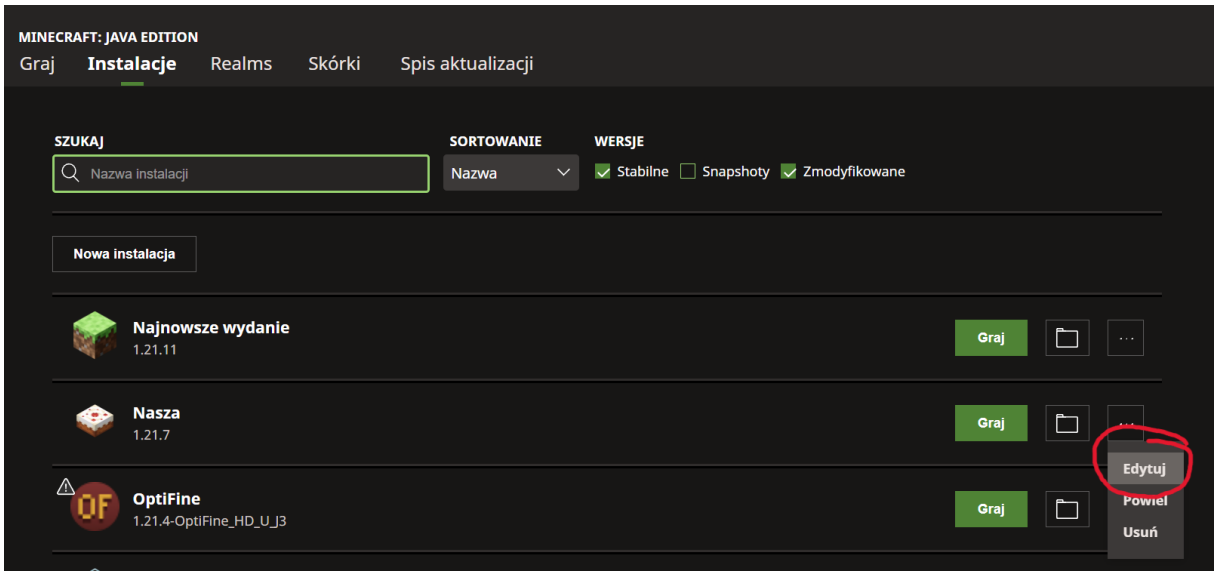
1.1 Open the Minecraft Launcher application



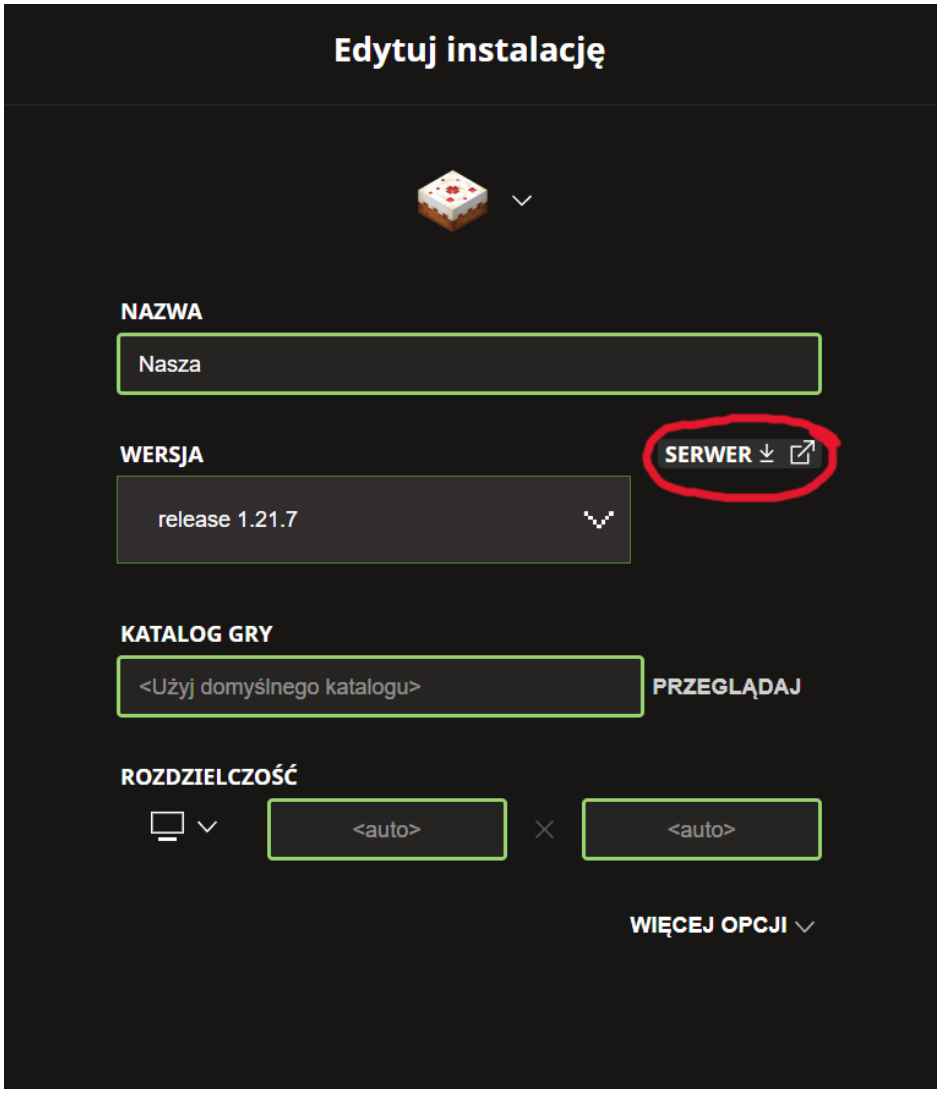
1.2 Go to the "Installations" tab

1.3 Select the installation of the game version on which you want to set up the server

1.4 Three-dot icon -> "Edit"



1.5 Click the "SERVER" link



2. Server installation

WARNING!!!

For the following steps to be completed successfully, you should check whether the latest version of Java software is installed on the device.

If it is not installed, the file should be downloaded from the official website and installed.

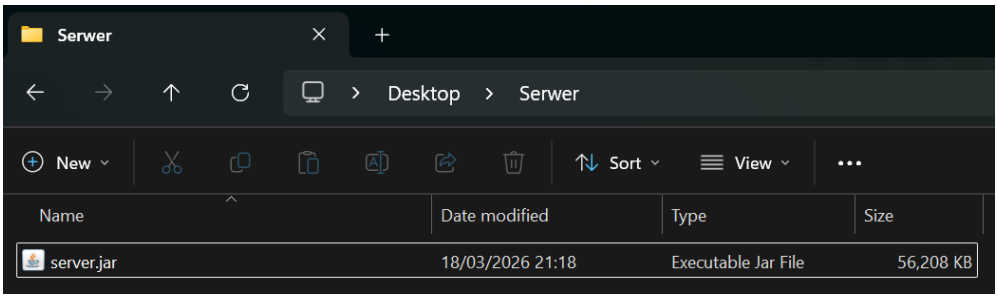
https://www.java.com/pl/download/

The following command in the command prompt allows you to check the version of the Java software

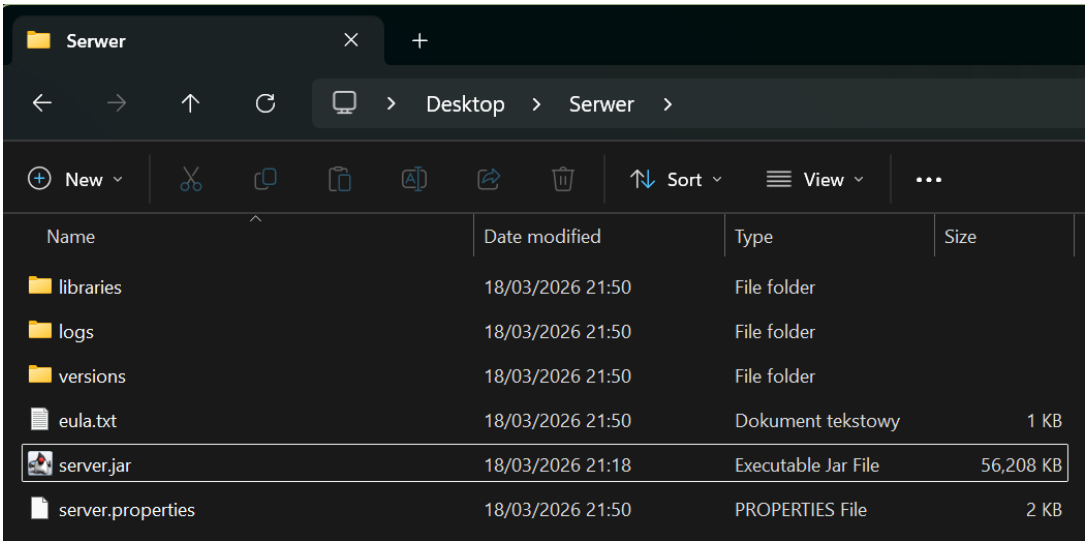
```
C:\Users\macie>Java -version
openjdk version "25.0.1" 2025-10-21 LTS
OpenJDK Runtime Environment (build 25.0.1+13-LTS)
OpenJDK 64-Bit Server VM (build 25.0.1+13-LTS, mixed mode, sharing)
```

The server installation proceeds as follows:

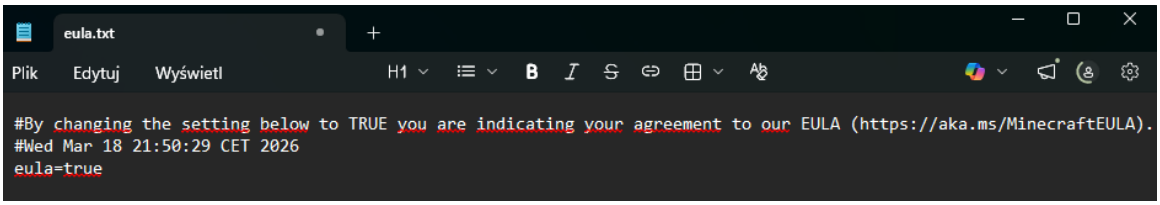
2.1. Create a new folder and move the server.jar file into it;



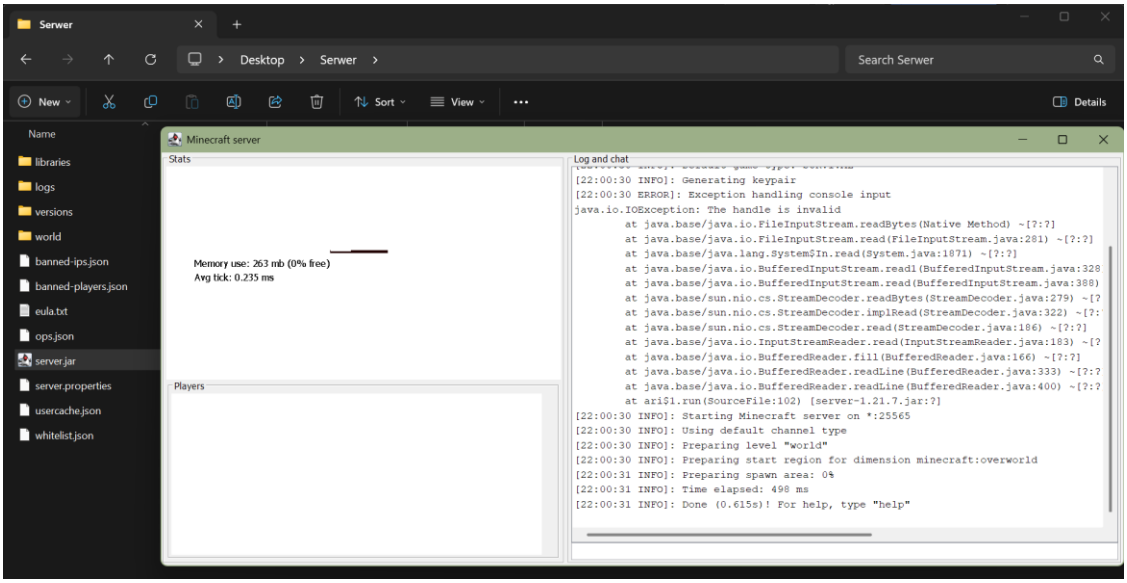
2.2. After opening the file, the following directories and files will be created;



2.3. To continue, open the eula.txt file, change "*eula=false" to "eula=true"* and save the file;



2.4. After restarting server.jar, the server will already be running locally on the device

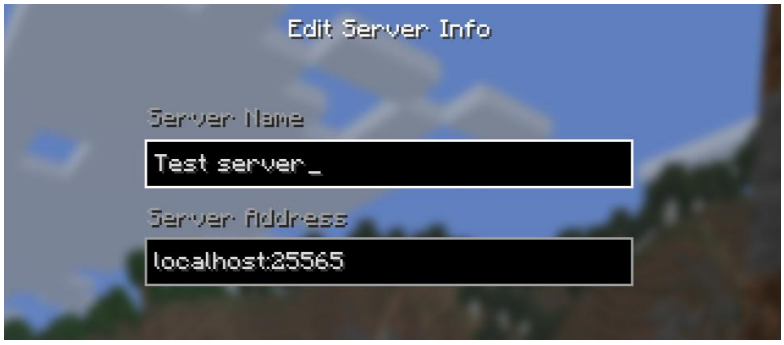


The server runs locally by default on port 25565.

2.5.

3. Connection test

To connect to the server from the client application, launch the game in the correct version (in this case version 1.21.7) and, in "Multiplayer" mode, enter localhost:[server port].



View from the client application after a successful connection

Log and chat

```
        at java.base/java.io.BufferedInputStream.read1(BufferedInputStream.java:328
        at java.base/java.io.BufferedInputStream.read(BufferedInputStream.java:388)
        at java.base/sun.nio.cs.StreamDecoder.readBytes(StreamDecoder.java:279) ~[?
        at java.base/sun.nio.cs.StreamDecoder.implRead(StreamDecoder.java:322) ~[?:
        at java.base/sun.nio.cs.StreamDecoder.read(StreamDecoder.java:186) ~[?:?]
        at java.base/java.io.InputStreamReader.read(InputStreamReader.java:183) ~[?
        at java.base/java.io.BufferedReader.fill(BufferedReader.java:166) ~[?:?]
        at java.base/java.io.BufferedReader.readLine(BufferedReader.java:333) ~[?:?
        at java.base/java.io.BufferedReader.readLine(BufferedReader.java:400) ~[?:?
        at ari$1.run(SourceFile:102) [server-1.21.7.jar:?]
[22:00:30 INFO]: Starting Minecraft server on *:25565
[22:00:30 INFO]: Using default channel type
[22:00:30 INFO]: Preparing level "world"
[22:00:30 INFO]: Preparing start region for dimension minecraft:overworld
[22:00:31 INFO]: Preparing spawn area: 0%
[22:00:31 INFO]: Time elapsed: 498 ms
[22:00:31 INFO]: Done (0.615s)! For help, type "help"
[22:01:31 INFO]: Server empty for 60 seconds, pausing
[22:04:21 INFO]: UUID of player [redacted] is [redacted]
[22:04:21 INFO]: [redacted][/127.0.0.1:44581] logged in with entity id 10 at (81.5,
[22:04:21 INFO]: [redacted] joined the game
[22:08:59 INFO]: [redacted] lost connection: Disconnected
[22:08:59 INFO]: [redacted] left the game
[22:09:59 INFO]: Server empty for 60 seconds, pausing
```

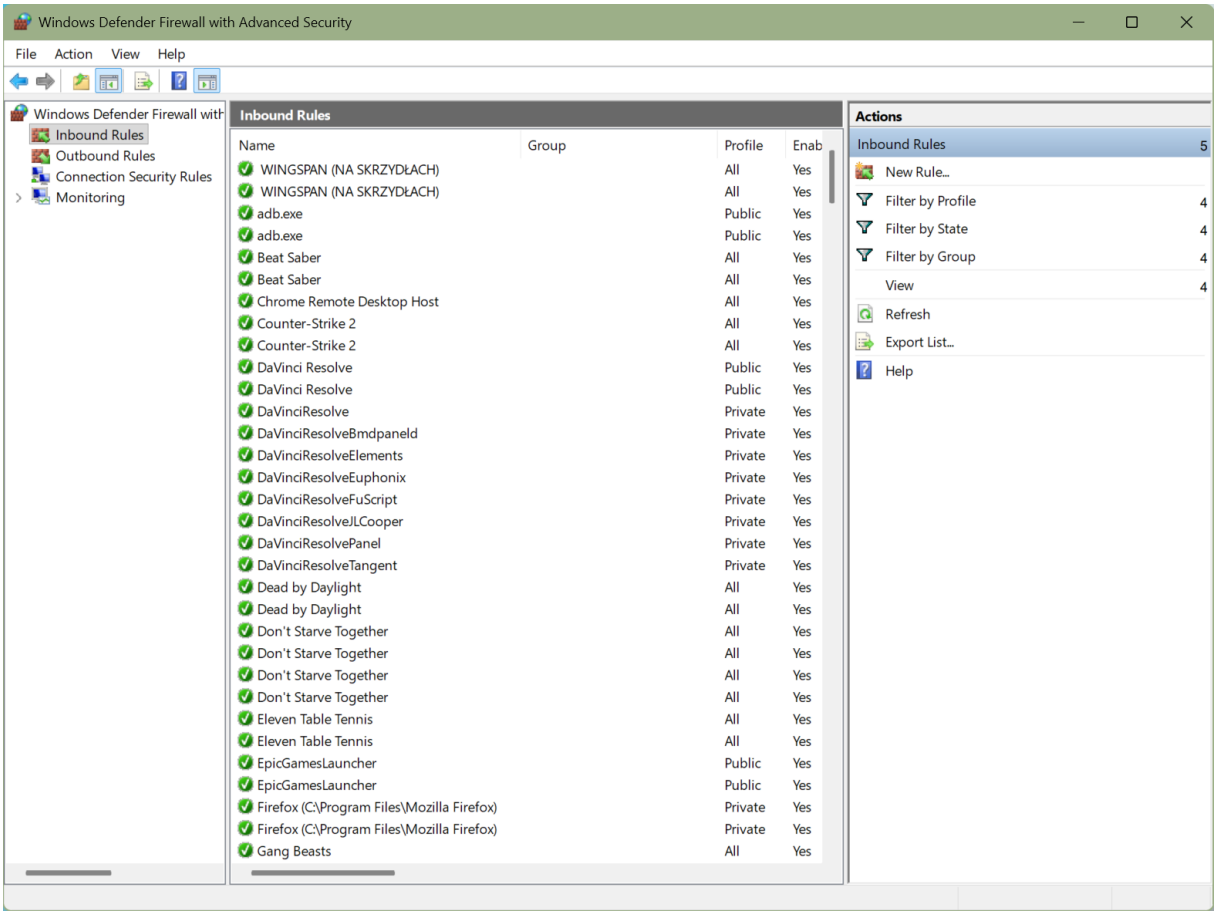
Server application log after a successful connection with the client

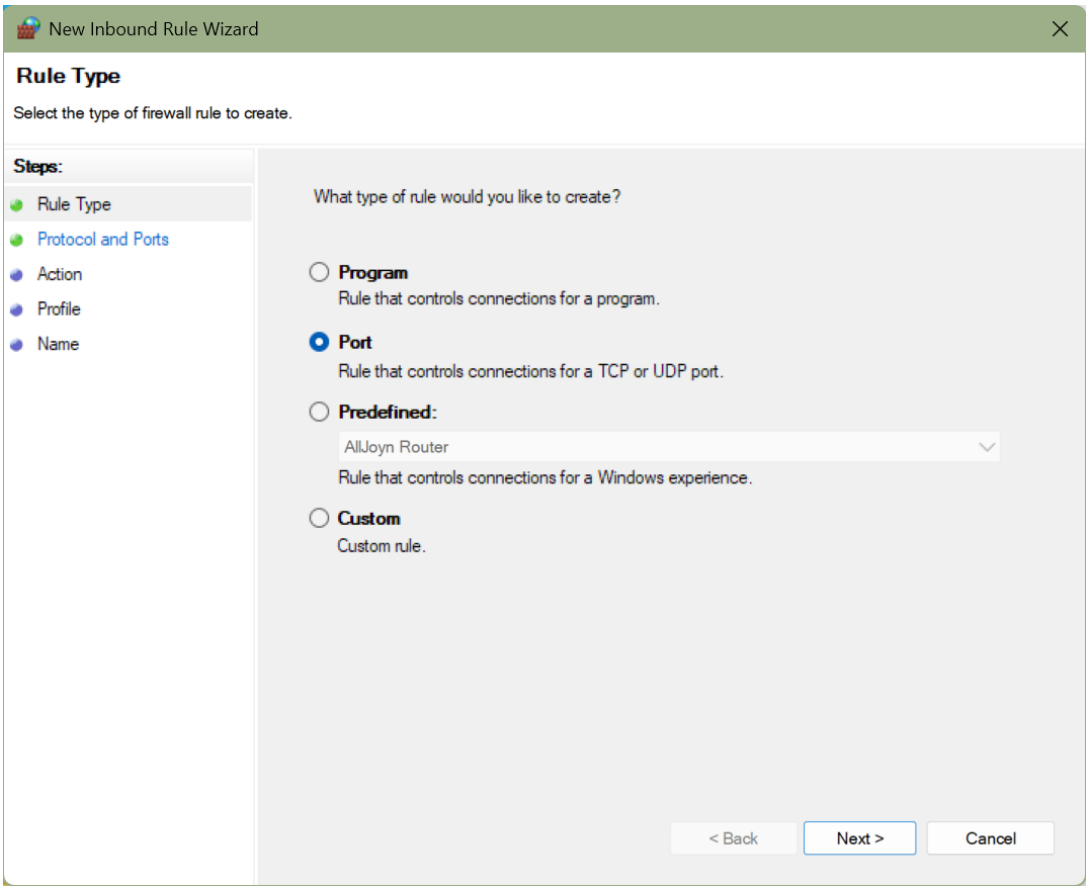
4. Configuration to allow devices outside the LAN to connect to the server

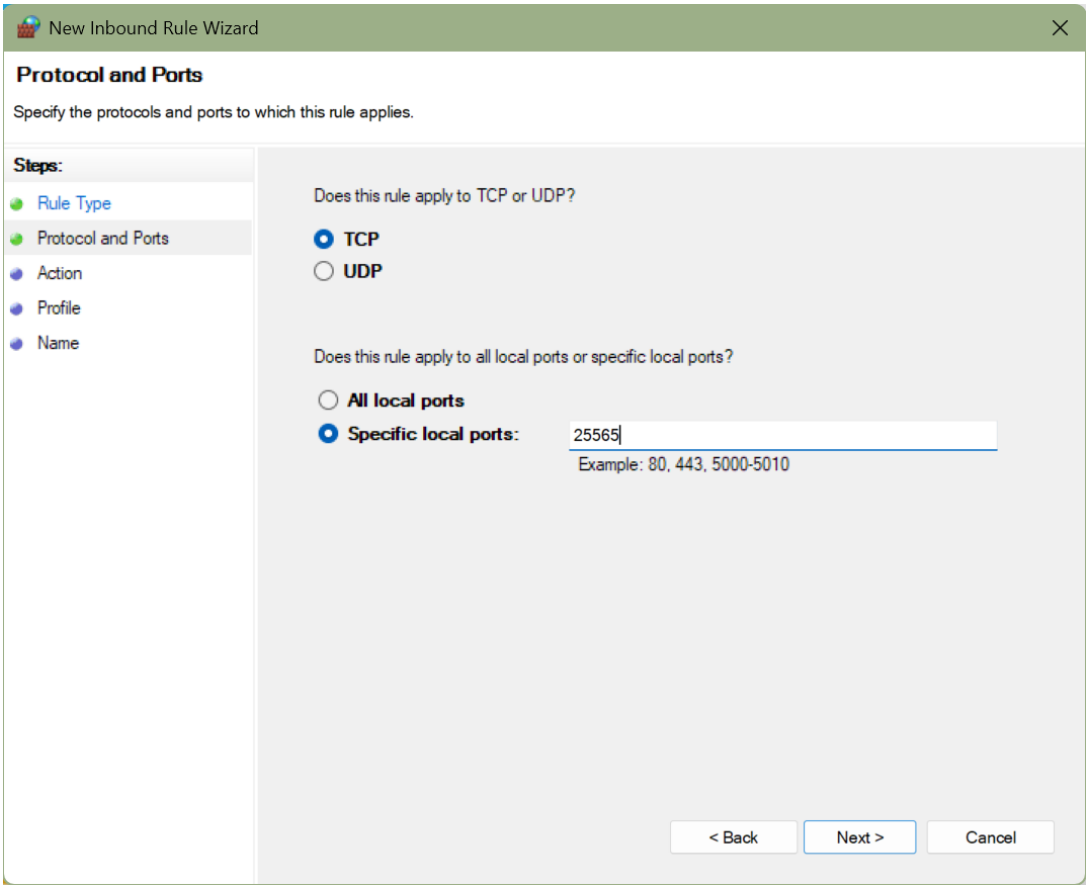
To allow players outside the local network to connect, the following steps must be performed:

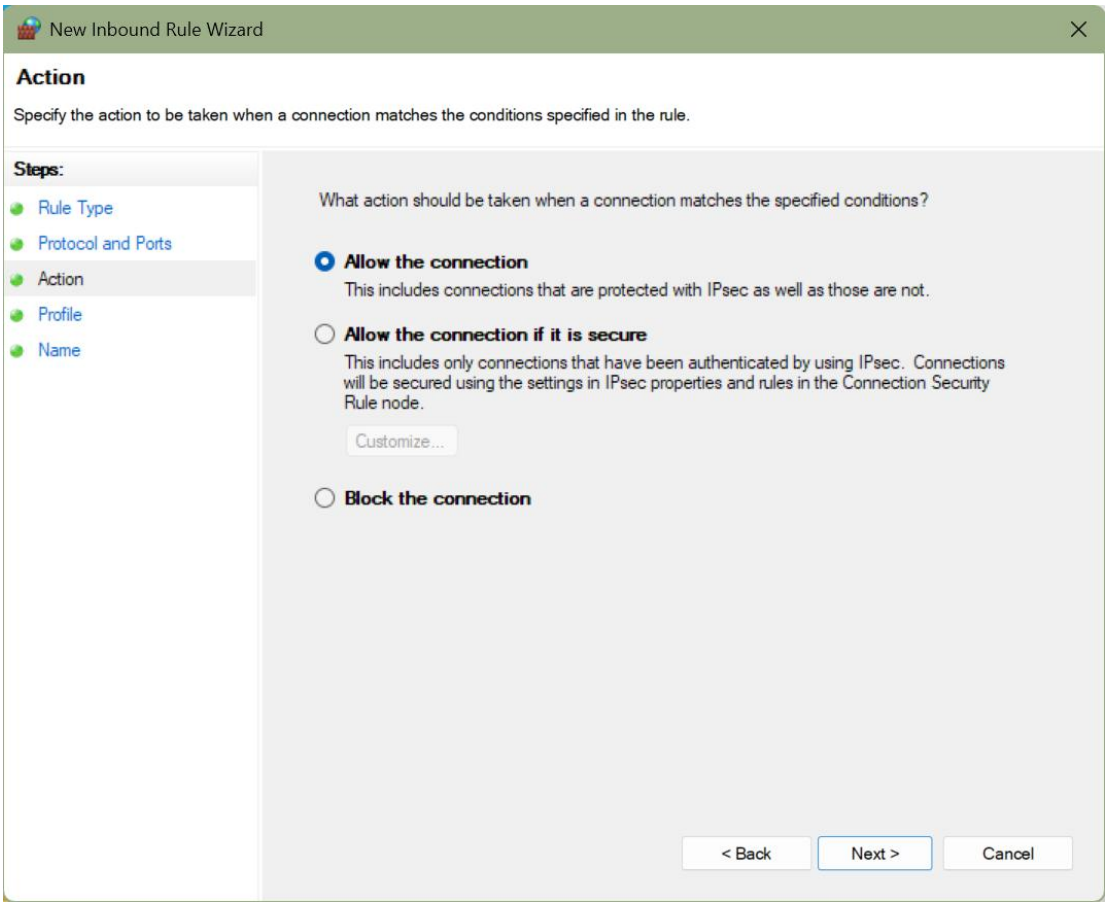
4.2. Configuring firewall rules

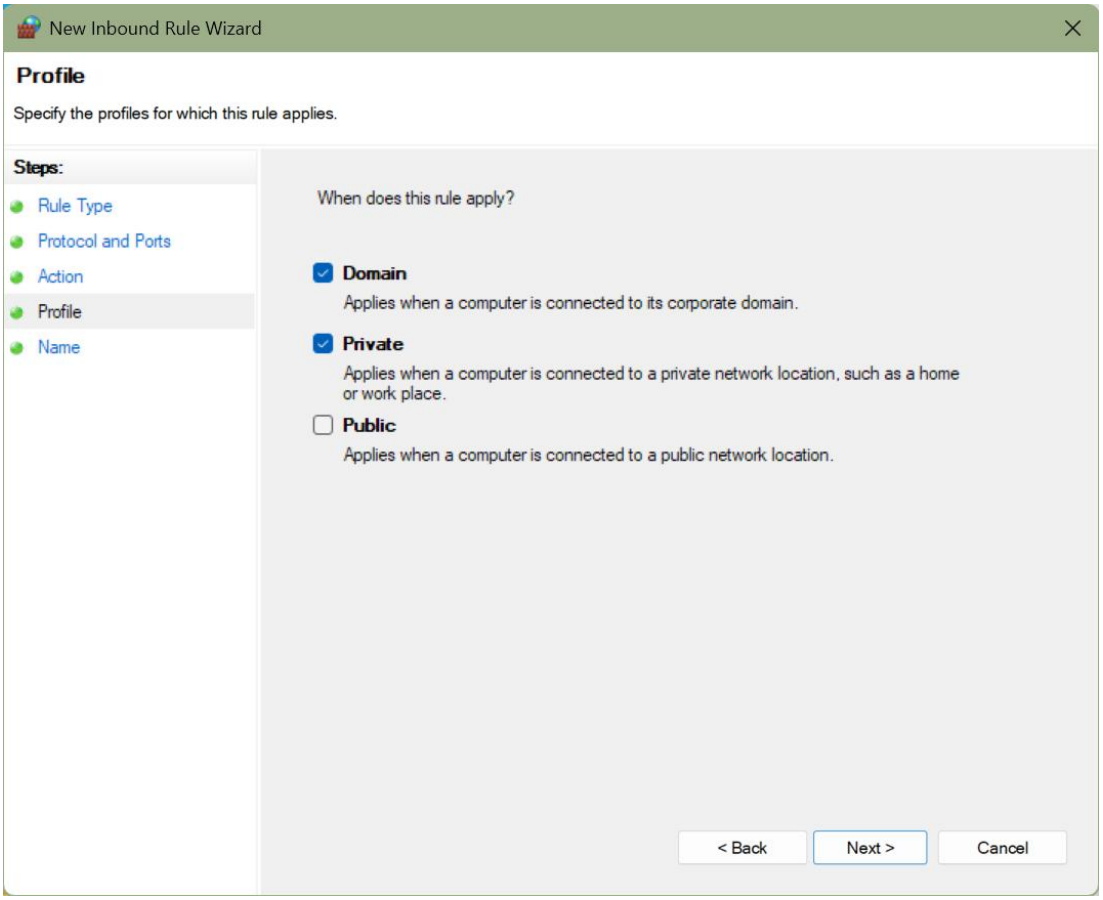
The screenshots below show the process of configuring inbound rules:

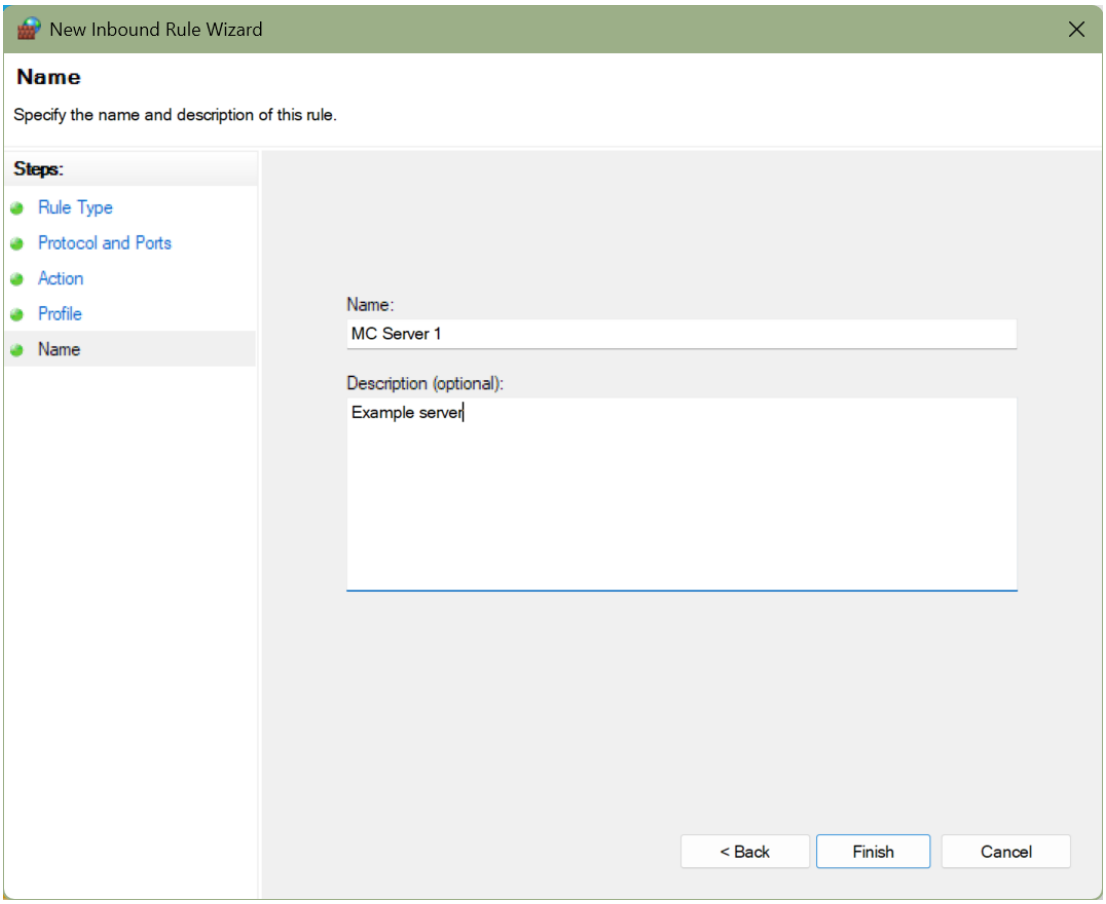


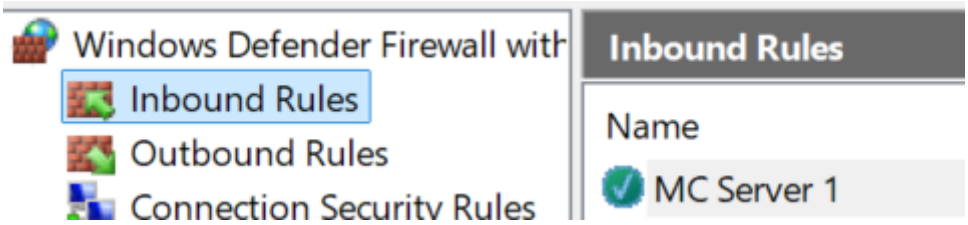

New Inbound Rule Wizard
Rule Type
Select the type of firewall rule to create.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
What type of rule would you like to create?
Program
Rule that controls connections for a program.
Port
Rule that controls connections for a TCP or UDP port.
Predefined:
AllJoyn Router
Rule that controls connections for a Windows experience.
Custom
Custom rule.
< Back
Next >
Cancel


New Inbound Rule Wizard
Protocol and Ports
Specify the protocols and ports to which this rule applies.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
Does this rule apply to TCP or UDP?
TCP
UDP
Does this rule apply to all local ports or specific local ports?
All local ports
Specific local ports:
25565
Example: 80, 443, 5000-5010
< Back
Next >
Cancel

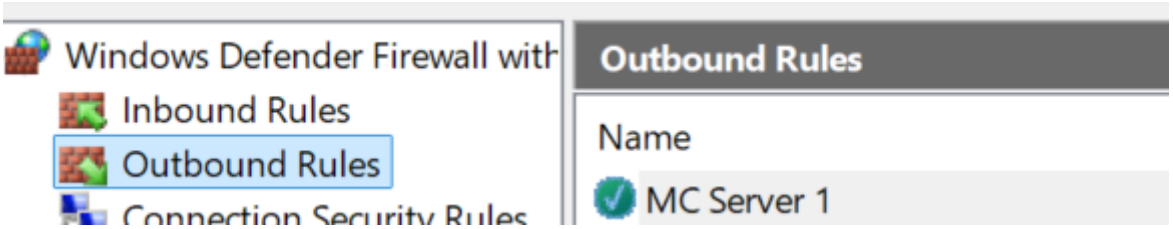

New Inbound Rule Wizard
Action
Specify the action to be taken when a connection matches the conditions specified in the rule.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
What action should be taken when a connection matches the specified conditions?
Allow the connection
This includes connections that are protected with IPsec as well as those are not.
Allow the connection if it is secure
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.
Customize...
Block the connection
< Back
Next >
Cancel


New Inbound Rule Wizard
Profile
Specify the profiles for which this rule applies.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
When does this rule apply?
Domain
Applies when a computer is connected to its corporate domain.
Private
Applies when a computer is connected to a private network location, such as a home or work place.
Public
Applies when a computer is connected to a public network location.
< Back
Next >
Cancel

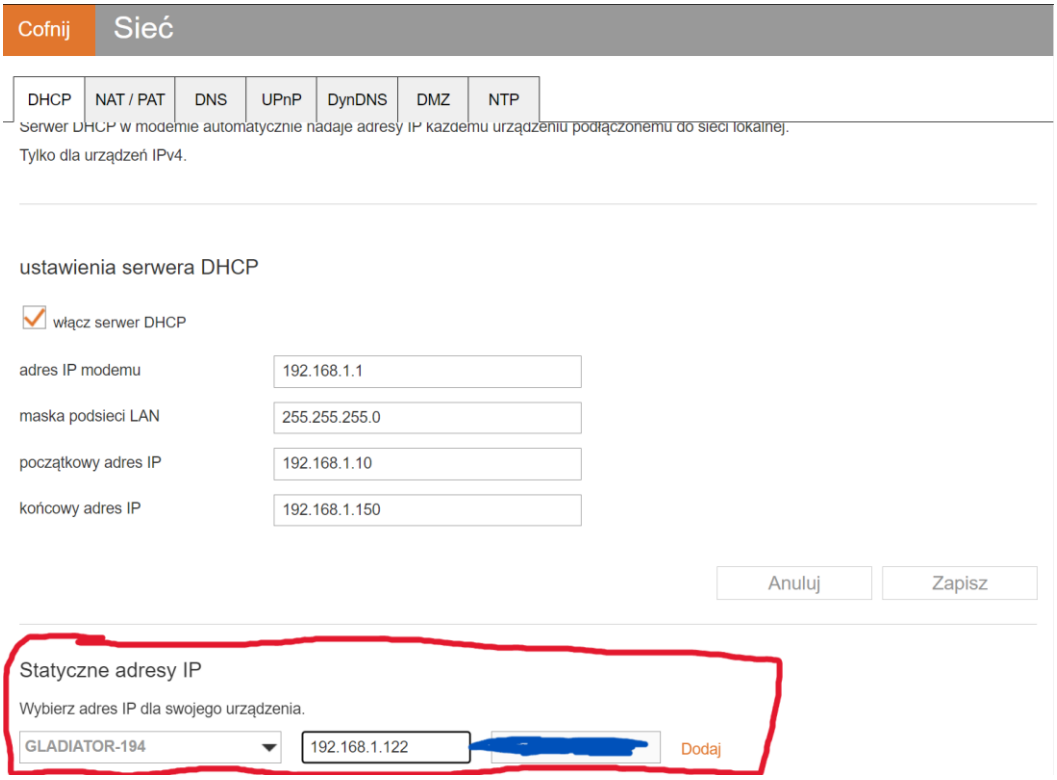




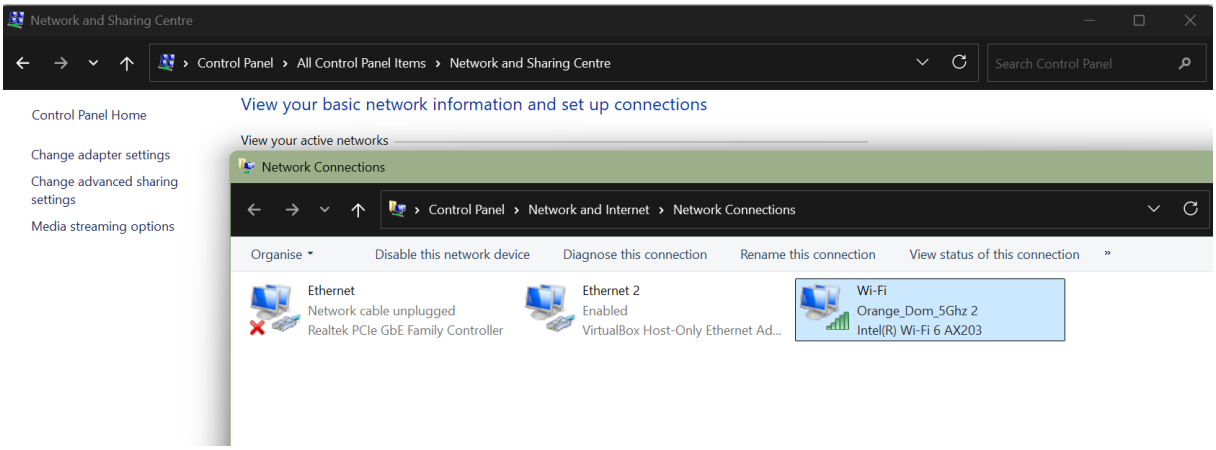
The outbound rule is configured in the same way.

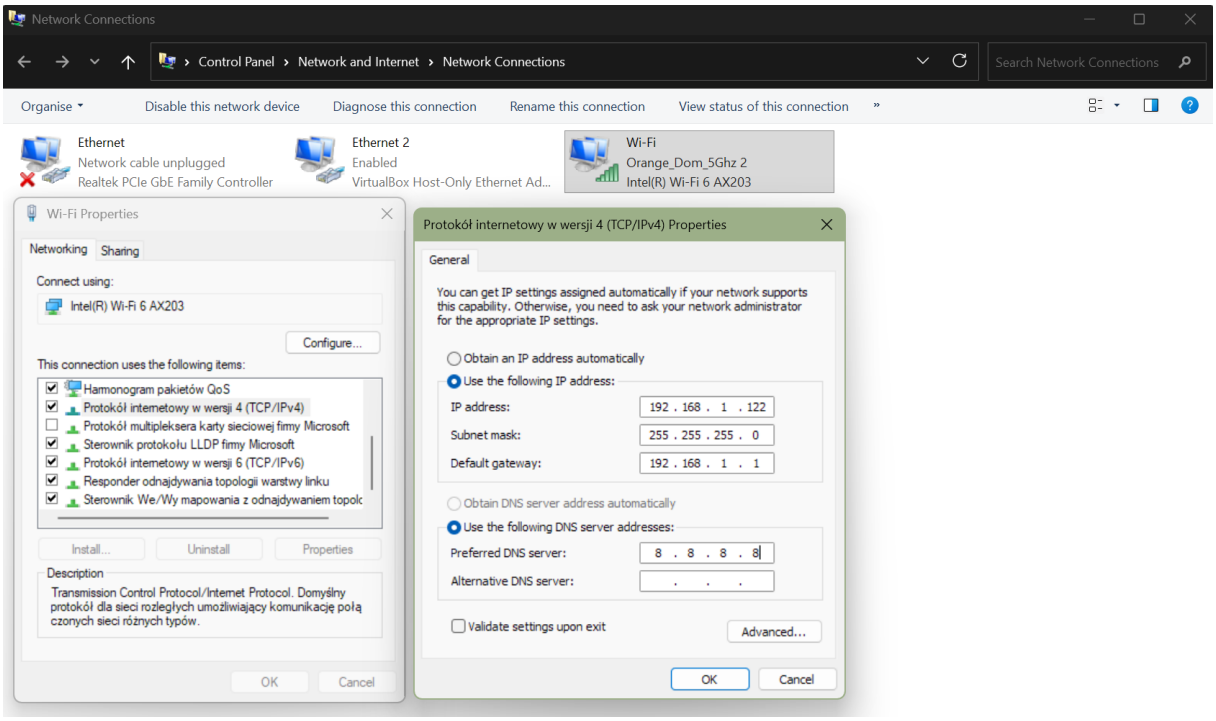


4.3. Configuring a static IP address for the device locally and on the router



Configuring a static address on the router.





Configuring a static IPv4 address on the server.

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::28ae:11ea:b51b:59ec%17
   IPv4 Address. . . . . . . . . . . : 192.168.1.122
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
```

Verifying the IP address using the ipconfig command.

4.4. In the server.properties file, the address of our server should be entered.

```
resource-pack=
resource-pack-id=
resource-pack-prompt=
resource-pack-sha1=
server-ip=192.168.1.122
server-port=25565
simulation-distance=10
spawn-monsters=true
spawn-protection=0
```

4.5. Configuring port forwarding on the router

Cofnij **Sieć**

DHCP | NAT / PAT | DNS | UPnP | DynDNS | DMZ | NTP

Reguły te umożliwiają komunikację z Internetu z określonym urządzeniem w sieci LAN. Możesz także zdefiniować porty, które będą używane podczas komunikacji.

Tylko dla urządzeń IPv4.

Reguły użytkownika

Uwaga! Upewnij się, że zapora sieciowa nie filtruje tych portów.

Tworzenie reguł na porcie 53 (DNS) jest zablokowane.

MC server | 25565 (np.: 1001) | 48494 (np.: 1000-2001) | TCP | GLADIATOR-... | Utwórz

status	aktywuj	aplikacja / usługa	port wewnętrzny	port zewnętrzny	protokół	urządzenie	
⚠	☐	Minecraft Server	25565	49494	TCP	192.168.1.112	🗑
⚠	☐	Nova plugin	38519	38519	TCP	192.168.1.112	🗑

FTP Server | 21 (np.: 1001) | 21 (np.: 1000-2001) | TCP | GLADIATOR-... | Utwórz

status	aktywuj	aplikacja / usługa	port wewnętrzny	port zewnętrzny	protokół	urządzenie	
⚠	☐	Minecraft Server	25565	49494	TCP	192.168.1.112	🗑
⚠	☐	Nova plugin	38519	38519	TCP	192.168.1.112	🗑
✓	☑	MC server	25565	48494	TCP	GLADIATOR-194	🗑

4.6. Checking the public IP address

Command Prompt

```
C:\Users\macie>nslookup myip.opendns.com resolver1.opendns.com
Server:  dns.sse.cisco.com
Address:  [redacted]222

Non-authoritative answer:
Name:    myip.opendns.com
Address:  83.[redacted]72
```

A client outside the local network must enter the server's public IP address and the external port.

For such a client, the server address should look as follows:

83.x.x.x72:48494

Conclusions

Based on the actions I carried out, I conclude that the Minecraft server was prepared and launched correctly. I managed not only to set it up locally, but also to configure it in such a way that it could be accessed from outside the LAN. To achieve this, I configured the system firewall, assigned a static IP address, forwarded the appropriate ports on the router, and verified the public IP address. The entire process confirmed that the server operates correctly and is ready to handle both local and external connections.

Sources

https://minecraft.wiki/w/Tutorial:Setting_up_a_Java_Edition_server